SREE MUTHUKUMARASWAMY COLLEGE

PROJECT TITLE

FitBuddy

AI fitness plan generator using Gemini models



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Better Health Brighter Future

FitBuddy – AI Fitness

Plan Generator using Gemini Models

Your Personal AI Fitness Coach



Project Introduction

FitBuddy – AI Fitness is a smart web application that generates personalized workout and nutrition plans using Google's Gemini AI models. It helps users achieve their fitness goals with AI-powered recommendations based on their health details, goals and preferences.

Objectives

- To build an AI-powered fitness plan generator using Gemini models.
- To provide personalized workout and diet plans based on user input.
- To make fitness guidance simple, smart and accessible to everyone.

Key Features

- Personalized workout plans (beginner, intermediate, advanced).
- Customized meal plans based on diet preferences.
- AI chat support for fitness queries.
- Progress tracking and daily motivation tips.

How It Works

User Input (Goal, Age, Weight, Diet, etc.) → Gemini AI Processes the Data → Personalized Plan Generated → User Follows & Tracks Progress

Technology Used

- **Frontend** : HTML, CSS, JavaScript (Streamlit)
- **Backend** : Python
- **AI Model** : Google Gemini Pro
- **Libraries** : google-generativeai, pandas, streamlit

System Design



Gemini AI Integration

- We use Google's Gemini Pro model to generate personalized fitness and nutrition plans.
- The user's input (goal, age, weight, diet, etc.) is sent to Gemini with a structured prompt.
- Gemini returns a detailed and customized plan in natural language format.

gemini_integration.py

```
import google.generativeai as genai

genai.configure(api_key="YOUR_GEMINI_API_KEY")

model = genai:GenerativeModel("gemini-pro")

response = model.generate_content(prompt)
print(response.text)
```

Gemini helps us get smart, accurate and personalized fitness suggestions.

Coding (Python + HTML)

app.py (Python - Streamlit)

```
import streamlit as st
import google.generativeai as genai
import os

genai.configure(api_key="YOUR_GEMINI_API_KEY")
model = genai.GenerativeModel("gemini-pro")

st.title("FitBuddy - AI Fitness")

name = st.text_input("What is your name?")
goal = st.selectbox("Your Goal",  ["Weight Loss",
  "Muscle Gain", "General Fitness"])

if st.button("Generate Plan"):
   prompt = f"Create a detailed workout and
      nutrition plan for a {goal} goal."
      response = model.generate_content(prompt)
   st.success(response.text)
)
```

index.html (Frontend - Simple UI)

```
<!DOCTYPE html>
<html>
<head>
  <title>FitBuddy</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>FitBuddy - AI Fitness</h1>
    <input type="text" id="name" placeholder="Your Name">
    <select id="goal">
      <option value="weight_loss">Weight Loss</option>
      <option value="muscle_gain">Muscle Gain</option>
      <option value="general">General Fitness</option>
      </select>
     <button onclick="generatePlan()">Generate Plan</bu:
    <div id="output"></div>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Sample Output



Final Output (Web Application Screenshot)



Advantages

- Personalized and accurate plans
- AI-powered smart recommendations
- Easy to use and accessible
- Helps build healthy habits

Limitations

- Depends on internet connection
- Requires valid API key
- May not replace professional trainers
- Limited to text-based output

Future Enhancements

- Add voice assistant
- Include progress tracking with graphs
- Mobile app version
- Integration with wearable devices

Conclusion

FitBuddy – AI Fitness combines the power of Gemini AI with modern web technologies to provide personalized workout and nutrition plans. It makes fitness guidance smarter, easier and more accessible, helping users live a healthier and happier life.

FitBuddy - AI Fitness Plan Generator using Gemini Models

1. Project Overview

FitBuddy is an intelligent, personalized fitness and nutrition planning application powered by Google's Gemini AI models. Traditional fitness applications offer static, pre-configured workout plans that fail to adapt to individual health goals, dietary restrictions, physical constraints, or real-time progress. FitBuddy solves this by dynamic real-time Generation of tailored workout routines and meal plans using advanced Prompt Engineering and Large Language Models (LLMs).

Key Features

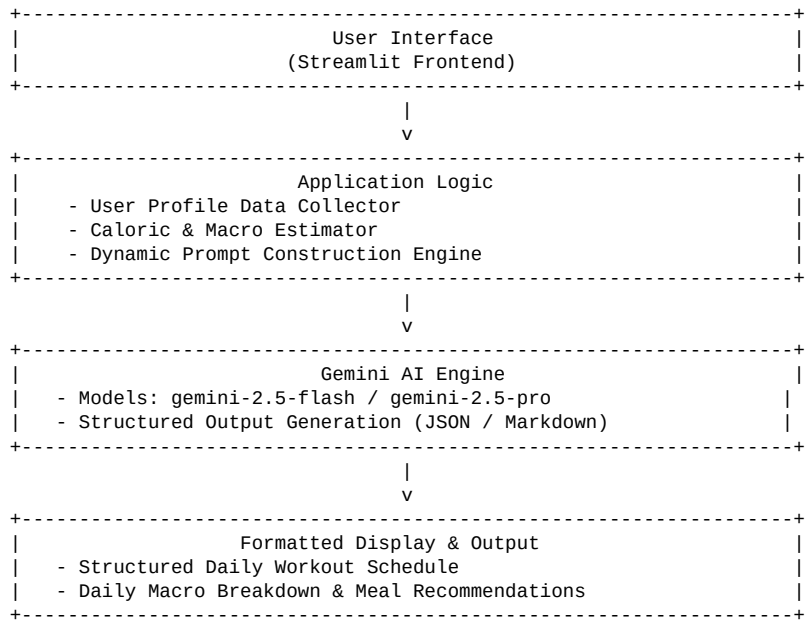
• Tailored Fitness Plans: Custom routines based on age, gender, fitness level, available equipment, and specific goals (e.g., muscle gain, weight loss, endurance).

• Diet & Nutrition Breakdown: Personal calorie and macronutrient (Carbs, Protein, Fats) calculation aligned with workout intensity.

• Health & Safety Constraints: Consideration of existing medical conditions, injuries, and dietary preferences (e.g., Vegan, Keto, Gluten-free).

• Interactive Dashboard: Built with Streamlit for a smooth, intuitive user experience.

2. System Architecture



3. Tech Stack

Component	Technology Used
Language	Python 3.10+
Frontend Framework	Streamlit
AI Model	Google Gemini API (google-genai SDK)
Model Version	gemini-2.5-flash
Environment Management	python-dotenv

4. Source Code Implementation

Create two files in your project folder: requirements.txt and app.py.

requirements.txt

```
streamlit>=1.35.0
google-genai>=0.1.1
python-dotenv>=1.0.1
```

app.py

```
import os
import streamlit as st
from google import genai
from google.genai import types
from dotenv import load_dotenv

# Load environment variables
load_dotenv()

# Page Setup
st.set_page_config(
    page_title="FitBuddy - AI Fitness Plan Generator",
    page_icon="■■■■■",
    layout="wide"
)

# Initialize Gemini Client
API_KEY = os.getenv("GEMINI_API_KEY")

if not API_KEY:
    st.error("■■ GEMINI_API_KEY not found! Please set your API key in the environment variables or .env file.")
    st.stop()

client = genai.Client(api_key=API_KEY)

# Header Section
st.title("■■■■■ FitBuddy: AI Fitness Plan Generator")
st.subheader("Powered by Google Gemini Models")
st.markdown("Enter your personal details, goals, and equipment below to receive a personalized workout and nutrition pr

st.divider()

# Input Form
with st.form("user_profile_form"):
    col1, col2, col3 = st.columns(3)

    with col1:
        age = st.number_input("Age", min_value=12, max_value=90, value=25)
        gender = st.selectbox("Gender", ["Male", "Female", "Non-binary", "Prefer not to say"])
        weight_kg = st.number_input("Weight (kg)", min_value=30.0, max_value=250.0, value=70.0)

    with col2:
        height_cm = st.number_input("Height (cm)", min_value=100.0, max_value=230.0, value=170.0)
        fitness_level = st.selectbox("Current Fitness Level", ["Beginner", "Intermediate", "Advanced"])
        primary_goal = st.selectbox("Primary Goal", ["Weight Loss", "Muscle Gain", "Endurance & Stamina", "General Heal

    with col3:
        workout_days = st.slider("Workout Days per Week", min_value=2, max_value=7, value=4)
        equipment = st.multiselect(
            "Available Equipment",
            ["Bodyweight Only", "Dumbbells", "Barbell", "Resistance Bands", "Kettlebell", "Gym Full Access"],
            default=["Bodyweight Only"]
        )
        dietary_pref = st.selectbox("Dietary Preference", ["No Preference", "Vegetarian", "Vegan", "Keto", "High Protei

    st.markdown("### Health & Safety (Optional)")
    injuries = st.text_input("Injuries or Medical Conditions (e.g., Lower back pain, Asthma, Knee issues)", placeholder

    submit_btn = st.form_submit_button("■ Generate My Fitness Plan", type="primary")

# Prompt Generation & API Call
if submit_btn:
    if not equipment:
        st.warning("Please select at least one available equipment option.")
    else:
        with st.spinner("Analyzing your profile and generating your custom fitness plan with Gemini..."):

            prompt = f"""
            You are an elite certified personal trainer and sports nutritionist.
```

```
Create a highly detailed, actionable, safe, and personalized weekly fitness and meal plan for the following

### USER PROFILE:
- Age: {age}
- Gender: {gender}
- Height: {height_cm} cm
- Weight: {weight_kg} kg
- Fitness Level: {fitness_level}
- Primary Goal: {primary_goal}
- Target Workout Frequency: {workout_days} days per week
- Available Equipment: {', '.join(equipment)}
- Dietary Preference: {dietary_pref}
- Existing Injuries/Conditions: {injuries if injuries else 'None'}

### OUTPUT INSTRUCTIONS:
Please structure your response clearly using Markdown formatting with the following exact sections:

1. **Executive Health Overview**
   - Estimated Daily Caloric Requirement (TDEE & Target Calories based on goal)
   - Recommended Macronutrient Split (Protein / Carbs / Fats in grams and percentages)

2. **Weekly Workout Routine ({workout_days} Days)**
   - Provide a day-by-day workout breakdown for {workout_days} training days and rest days.
   - Include specific exercise names, Sets, Reps, and Rest periods.
   - Ensure selected exercises strictly align with available equipment: {', '.join(equipment)}.
   - Account for injuries: {injuries if injuries else 'None'}.

3. **Daily Nutrition & Sample Meal Plan**
   - Sample Breakfast, Lunch, Dinner, and Snack ideas tailored to dietary preference ({dietary_pref}).
   - Hydration and supplementation tips.

4. **Safety & Coaching Tips**
   - Form advice and injury avoidance.
   - Progressive overload strategy for long-term progress.
"""

try:
    response = client.models.generate_content(
        model="gemini-2.5-flash",
        contents=prompt,
        config=types.GenerateContentConfig(
            temperature=0.7,
            top_p=0.9,
        )
    )

    st.success("■ Your Personalized FitBuddy Plan is Ready!")
    st.markdown(response.text)

except Exception as e:
    st.error(f"An error occurred while generating your plan: {str(e)}")
```

5. Project Setup & Execution

Step 1: Clone or Create Project Directory

```
mkdir fitbuddy-gemini
cd fitbuddy-gemini
```

Step 2: Set Up Virtual Environment

```
python -m venv venv

# On Windows:
venv\\Scripts\\activate

# On Linux/macOS:
source venv/bin/activate
```

Step 3: Install Dependencies

```
pip install -r requirements.txt
```

Step 4: Configure API Key

```
Create a .env file in the project root directory:

GEMINI_API_KEY="YOUR_ACTUAL_GEMINI_API_KEY_HERE"
```

Step 5: Run the Streamlit Application

```
streamlit run app.py
```

The application will launch automatically in your browser at:

```
http://localhost:8501
```

6. Test Plan & Expected Results

Test Case	Inputs	Expected Output
TC-01: Beginner Home Routine	Weight: 70kg, Goal: Weight Loss, Equip: Bodyweight Only	Calorie deficit plan, low-impact bodyweight exercises (pushups, squats), 3-day sp
TC-02: Advanced Gym Lifting	Weight: 85kg, Goal: Muscle Gain, Equip: Full Gym Access	Hypertrophy workout split (Push/Pull/Legs), high protein diet plan, progressive ov
TC-03: Injury Safety Constraint	Goal: General Health, Injury: Lower Back Pain	Routine avoids heavy deadlifts or spinal compression moves, suggesting low-imp

7. Future Enhancements

• Computer Vision Form Check: Integrate Gemini Multimodal capabilities to analyze video clips or images of exercise form.

• Dynamic Progress Tracking: Allow users to log completed workouts and adjust target calories dynamically over time.

• PDF Export: Enable one-click downloading of generated weekly plans in PDF format.